

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
8	(a)			Set up the apparatus with e.g. 100 turns on the primary and 20 turns on the secondary (1) Set the primary voltage to a {suitable value / 3 V / 6 V} or keep the primary voltage constant (1) Measure the secondary voltage (1) Repeat with different numbers of <u>secondary</u> turns [keeping the primary turns constant] (1)	4			4		4
	(b)			a.c. produces an {alternating / changing} magnetic field (1) which is required to <u>induce</u> {voltage / current} in the secondary coil (1)	2			2		
	(c)			They increase the voltage (1) reducing {current / heating / energy loss} [in cables] (1) Don't accept stop heating effect / no energy loss	2			2		
	(d)			Substitution: $\frac{230}{11.5} = \frac{600}{N_2}$ (1) Rearrangement: $N_2 = \frac{600}{230} \times 11.5$ (1) = 30 (1)	1	1 1		3	3	
				Question 8 total	9	2	0	11	3	4